

Integral cubic torsion and Frobenius in genus two

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Abstract

Let T be a symplectic $\mathbb{Z}_2$ -lattice of rank four and let L be its closed-surface cubic Lie quotient. For a nondegenerate symmetric monodromy pairing B , we compute the two-torsion in $\text{coker}(\rho_3(\tau_B) - 1)$ and its normaliser action. For primitive B , the module is canonically the free cubic Lie module of the rank-two quotient by the vanishing-cycle Lagrangian, reduced modulo two. For even B , we give a complete fixed-dimension formula for every graph Frobenius action, including dimensions 2, 3, 4, 5, 6, 8, 10. For the thick theta pairing we also determine the full two-primary fixed groups when integral graph Frobenius is $\pm I$. Explicit curves over $\mathbb{Q}_5$ with identical metric graphs, graph action and ordinary component groups have different cubic invariants. The proof combines small integral weight maps, a Bockstein sequence and explicit finite row reductions. Two separate exact implementations cover twenty-four curves and a direct comparison with the classical Steinberg module. No novelty is claimed for the lattice or its reduced representation.

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1 Introduction

Rational descriptions of the unipotent fundamental group of a semistable curve are strongly governed by the geometry of its reduction graph. Integral structures can retain additional finite information that disappears after tensoring with $\mathbb{Q}_2$. This paper exhibits such a phenomenon in the cubic quotient attached to a genus-two curve.

Let T be the 2-adic Tate lattice in the totally toric genus-two setting, and let $A \subset T$ denote the vanishing-cycle Lagrangian. The cubic lattice considered below is the closed-surface degree-three Lie quotient. For tame inertia with monodromy pairing B , the central object is the subgroup annihilated by two in the cubic coinvariants,

$$D_B = (\text{coker}(\rho_3(\tau_B) - 1))[2].$$

The behaviour of D_B separates sharply according to whether the pairing is primitive or even at two.

In the primitive case, Theorem 2.2 gives a canonical normaliser-equivariant identification

$$D_B \simeq \text{Lie}_3((T/A) \otimes \mathbb{F}_2).$$

The resulting Frobenius action is exactly the graph action modulo two and is independent of the choice of complementary Lagrangian and of the integral extension block. For a rank-two quotient, D_B has dimension two.

The even case behaves differently. Theorem 2.3 shows that D_B has dimension ten. When graph Frobenius is trivial modulo two, the integral extension block remains visible and the fixed dimension is

$$\dim_{\mathbb{F}_2} D_B^F \in \{10, 8, 6\}.$$

More precisely, the three values are distinguished by whether the reduced extension block is zero, equal to the reduced primitive monodromy pairing, or neither. This information is not encoded by the metric graph and its Frobenius action alone.

The arithmetic significance is made explicit in Section 8. Theorem 8.1 constructs eight genus-two curves over $\mathbb{Q}_5$ with the same split metric theta graph of edge lengths $(2, 2, 2)$, the same trivial graph Frobenius action, the same geometric component group $\mathbb{Z}/2 \oplus \mathbb{Z}/6$, and the same ordinary Tamagawa number, while their cubic fixed dimensions are 10, 8 and 6. In this sense the cubic integral invariant detects extension data invisible to the usual graph-level invariants.

We also give a complete formula for nontrivial graph Frobenius: an involution has fixed dimension five or three/four, according to the primitive form and the extension, while an order-three graph action has fixed dimension two. For the thick theta pairing we determine the full two-primary fixed groups when integral graph Frobenius is I_2 or $-I_2$, including all eight unramified quadratic twists of the split family. Further explicit pairs have the same graph with action and ordinary component data, but cubic fixed dimensions five versus three and five versus four.

The proof is elementary at the lattice level. An explicit integral basis gives a four-step weight filtration. Three small weight maps control the kernel and reduction modulo two, and a Bockstein exact sequence converts the mod-two kernel calculation into the required integral two-torsion. A standalone exact verifier accompanies the paper and independently checks the displayed polynomial identities, finite normaliser cases, row-space calculations and discriminants used in the examples.

The relationship with existing work is discussed in Section 9. The surface Lie presentation is classical, and Berry recently constructed the same integral tensor lattice in a broader hyperelliptic cubic-Chabauty setting. The contribution here begins with the normaliser-equivariant calculation of its integral two-torsion and the extension-dependent Frobenius phenomenon.

2 Main results and setup

Let $T = A \oplus \Lambda$ be free of rank four over $\mathbb{Z}_2$, with a symplectic basis e_0, e_1, e_2, e_3 such that $A = \langle e_0, e_1 \rangle$ is Lagrangian and $\Lambda = \langle e_2, e_3 \rangle$. Write

$$\theta = e_0 \wedge e_2 + e_1 \wedge e_3, \quad j(x \wedge y \wedge z) = x \otimes (y \wedge z) + y \otimes (z \wedge x) + z \otimes (x \wedge y).$$

The lattice studied here is

$$L = \frac{T \otimes \wedge^2 T}{T \otimes \mathbb{Z}_2 \theta + j(\wedge^3 T)}. \quad (1)$$

It is the cubic quotient obtained from the free Lie algebra by the surface relation. This presentation and its use as an integral lattice in the rational cubic fundamental-group quotient are established background; see Labute, Filip and Berry [1, 2, 3]. We give an explicit integral basis below.

For $B = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$ with $\det B \neq 0$, put

$$\tau_B = \begin{pmatrix} I_2 & B \\ 0 & I_2 \end{pmatrix}, \quad \Delta_B = \rho_3(\tau_B) - I_L, \quad D_B = (\text{coker } \Delta_B)[2].$$

Here $M[2]$ denotes the subgroup annihilated by two, not the whole two-primary torsion subgroup. Write $B = 2^s B_0$, where $s \geq 0$ and B_0 has at least one unit entry. A bar denotes reduction modulo two.

Consider an integral normaliser of the form

$$F = \tau_H \begin{pmatrix} qU^{-t} & 0 \\ 0 & U \end{pmatrix}, \quad H = H^t, \quad U \in \mathrm{GL}_2(\mathbb{Z}_2), \quad U^t B U = B, \quad (2)$$

where q is a positive odd integer. It satisfies $F\tau_B F^{-1} = \tau_B^q$. All actions on L are induced by the tensor presentation (1).

Proposition 2.1. *The lattice L is free of rank sixteen. For every nondegenerate B , the kernel $\ker \Delta_B$ has rank six.*

Theorem 2.2 (Primitive pairing). *Suppose $s = 0$. There is a canonical normaliser-equivariant isomorphism*

$$\mathrm{Lie}_3((T/A) \otimes \mathbb{F}_2) \xrightarrow{\sim} D_B. \quad (3)$$

For a rank-two vector space V , $\mathrm{Lie}_3(V) = V \otimes \wedge^2 V$. Consequently D_B has dimension two and the action of F is $\bar{U}$. The isomorphism depends on A and the monodromy action, but not on a complementary Lagrangian or on the choice of an odd-unit tame generator.

Theorem 2.3 (Even pairing). *Suppose $s \geq 1$. Then D_B has dimension ten. If in addition $\bar{U} = I_2$, then*

$$\dim_{\mathbb{F}_2} D_B^F = \begin{cases} 10, & \bar{H} = 0, \\ 8, & \bar{H} = \bar{B}_0, \\ 6, & \bar{H} \neq 0, \bar{H} \neq \bar{B}_0. \end{cases} \quad (4)$$

Over $\mathbb{F}_2$ a rank-two determinant line has a unique nonzero element. Thus the identification of $V \otimes \wedge^2 V$ with V used in Theorem 2.2 is natural. In particular, the primitive fixed dimension is $\dim \ker(\bar{U} - I_2)$.

Corollary 2.4. *Let C be a genus-two curve with totally toric semistable reduction over a local field of odd residue characteristic. Let B be the pairing on the integral cycle lattice of its geometric reduction graph and use (1) with $T = T_2 \mathrm{Jac}(C)$. If B is primitive at two, its cubic coinvariant two-torsion is the graph module modulo two. If B is even, the integral extension data can affect this module's Frobenius invariants even when the metric graph and its Frobenius action are fixed.*

Section 6 extends the even formula to every graph action. Section 7 computes the full two-primary group for the thick theta pairing. The arithmetic construction and examples are given in Section 8; Appendix D verifies the model hypotheses and root/graph basis compatibility. The statements are local and concern the specified integral cubic lattice; no global Chabauty or Selmer-dimension conclusion is asserted.

3 Integral basis and weight maps

For convenience write $b_{i,jk} = e_i \otimes (e_j \wedge e_k)$, with $j < k$. First eliminate $b_{i,13}$ using $b_{i,13} = -b_{i,02}$. The four remaining Jacobi relations eliminate

$$\begin{aligned} b_{0;02} &= -b_{1;03} + b_{3;01}, & b_{0;12} &= b_{1;02} - b_{2;01}, \\ b_{0;23} &= b_{2;03} - b_{3;02}, & b_{1;23} &= -b_{2;02} - b_{3;12}. \end{aligned}$$

Every elimination has a unit coefficient and eliminates a different coordinate. This proves freeness and saturation over $\mathbb{Z}$, as well as over $\mathbb{Z}_2$.

Count the number of letters belonging to the complementary span of e_2, e_3 . The resulting direct sum $L = V_0 \oplus V_1 \oplus V_2 \oplus V_3$ has the following ordered bases:

Weight	Ordered basis	Rank
0	$b_{0;01}, b_{1;01}$	2
1	$b_{0;03}, b_{1;02}, b_{1;03}, b_{1;12}, b_{2;01}, b_{3;01}$	6
2	$b_{2;02}, b_{2;03}, b_{2;12}, b_{3;02}, b_{3;03}, b_{3;12}$	6
3	$b_{2;23}, b_{3;23}$	2

Although the splitting uses a complement, the increasing filtration $\mathcal{F}_i = \bigoplus_{k \leq i} V_k$ is intrinsic to A : it is induced by tensors with at least $3 - i$ letters in A . The surface and Jacobi relations respect this filtration. In particular,

$$L/\mathcal{F}_2 \simeq \text{Lie}_3(T/A). \quad (5)$$

Let $n_B : T \rightarrow T$ send $e_2 \mapsto ae_0 + be_1$, $e_3 \mapsto be_0 + ce_1$, and kill A . Its induced derivation N_B on L lowers weight by one. In the displayed bases its three blocks are

$$M_B : V_1 \rightarrow V_0, \quad M_B = \begin{pmatrix} c & 0 & 0 & 0 & a & b \\ 0 & b & c & -a & b & c \end{pmatrix}, \quad (6)$$

$$C_B : V_2 \rightarrow V_1, \quad C_B = \begin{pmatrix} 0 & a & 0 & 0 & b & 0 \\ b & 0 & a & c & 0 & b \\ -a & b & 0 & -b & c & 0 \\ 0 & 0 & b & 0 & 0 & c \\ b & c & -2a & 0 & 0 & -b \\ a & 0 & 0 & 2b & c & -a \end{pmatrix}, \quad (7)$$

$$E_B : V_3 \rightarrow V_2, \quad E_B = \begin{pmatrix} -3b & -c \\ 2a & b \\ -c & 0 \\ -a & -3b \\ 0 & a \\ -b & -2c \end{pmatrix}. \quad (8)$$

For example, these formulas follow by applying the derivation to each $b_{i;jk}$ and then using the eight eliminations above. They do not involve division by two.

Over $\mathbb{Q}_2$ one has $\rho_3(\tau_B) = \exp(N_B)$, because $\tau_B = \exp(n_B)$ and the construction is functorial. Since $N_B^4 = 0$, this gives the integral polynomial identity

$$6\Delta_B = 6N_B + 3N_B^2 + N_B^3. \quad (9)$$

We use this equality over $\mathbb{Q}_2$ to identify kernels, and do not divide by two in an integral change of basis.

Lemma 3.1. *For nondegenerate $B = 2^s B_0$,*

$$\ker \Delta_B = V_0 \oplus \ker M_{B_0}.$$

Moreover, M_{B_0} is surjective over $\mathbb{Z}_2$ and E_{B_0} has a unit rank-two minor. The reduction of the integral kernel is therefore

$$K := \overline{\ker \Delta_B} = \bar{V}_0 \oplus \ker \bar{M}_{B_0}, \quad \dim K = 6. \quad (10)$$

Proof. Expansion of (7) gives

$$\det C_B = 4b^6 - 12ab^4c + 12a^2b^2c^2 - 4a^3c^3 = -4(ac - b^2)^3.$$

Thus C_B is invertible over $\mathbb{Q}_2$. If a is a unit, M_B has a minor $\pm a^2$ in columns 4, 5, and E_B has a minor $-a^2$ in rows 4, 5. If c is a unit, use columns 1, 3 of M_B and rows 1, 3 of E_B , with minors $\pm c^2$. If only b is a unit, columns 2, 6 of M_B have minor $-b^2$, and rows 1, 2 of E_B have minor $-3b^2 + 2ac$, again a unit. Apply this argument to B_0 .

The weight blocks give $\ker N_B = V_0 \oplus \ker M_{B_0}$ over $\mathbb{Q}_2$. In characteristic zero, $\exp N_B - I = N_B(1 + N_B/2 + N_B^2/6)$ and the parenthesised factor is invertible. Intersecting with the free lattice L proves the integral kernel formula. Surjectivity of M_{B_0} makes reduction of its kernel exact, proving (10). $\square$

Proof of Proposition 2.1. The explicit eliminations above give a free rank-sixteen lattice. Lemma 3.1 identifies $\ker \Delta_B$ with $V_0 \oplus \ker M_{B_0}$; since $M_{B_0} : V_1 \rightarrow V_0$ is surjective, this kernel has rank $2 + (6 - 2) = 6$. $\square$

4 Reduction modulo two and the canonical map

Lemma 4.1. *For every nonzero symmetric $\bar{B}$, the ranks of $\bar{M}_B, \bar{C}_B, \bar{E}_B$ are 2, 4, 2, respectively. Moreover,*

$$\bar{\Delta}_B^2 = 0, \quad \text{im } \bar{\Delta}_B = \ker \bar{\Delta}_B, \quad \dim \ker \bar{\Delta}_B = 8.$$

Proof. The unit minors above give the outer ranks. Since $n_B^2 = 0$ on T and cross terms in the square of its tensor derivation have coefficient two, $\bar{N}_B^2 = 0$. In particular $\bar{C}_B \bar{E}_B = 0$, so $\text{rank } \bar{C}_B \leq 4$.

Every nonzero symmetric bilinear form in dimension two over $\mathbb{F}_2$ is congruent to one of the following three forms. For each, the indicated 4×4 minor of (7) is odd. Row and column numbers start at one.

(a, b, c)	Rows	Columns	Determinant
$(1, 0, 0)$	1, 2, 3, 6	1, 2, 3, 6	1
$(1, 0, 1)$	1, 2, 3, 4	1, 2, 3, 6	-1
$(0, 1, 0)$	1, 2, 3, 4	1, 2, 3, 5	-1

The induced changes of basis of A and its dual complement preserve the weight-block ranks. Hence $\text{rank } \bar{C}_B = 4$ for every nonzero $\bar{B}$.

The leading weight blocks of $\bar{\Delta}_B$ are $\bar{M}_B, \bar{C}_B, \bar{E}_B$, so $\text{rank } \bar{\Delta}_B \geq 2 + 4 + 2 = 8$. On the other hand $\bar{\tau}_B^2 = I_T$, giving $\bar{\Delta}_B^2 = 0$. On a sixteen-dimensional space this bounds its rank by eight, proving all assertions. $\square$

For a map Δ of free $\mathbb{Z}_2$ -modules, reduction of its kernel-cokernel sequence gives the Bockstein exact sequence

$$0 \longrightarrow (\ker \Delta)/2 \longrightarrow \ker \bar{\Delta} \xrightarrow{\beta} (\text{coker } \Delta)[2] \longrightarrow 0. \quad (11)$$

Explicitly, if $\bar{x} \in \ker \bar{\Delta}$ and x is a lift, then $\beta(\bar{x}) = [\Delta x/2]$. Changing the lift adds an element of $\text{im } \Delta$. If the image under β is zero, subtracting twice a suitable vector yields a lift in $\ker \Delta$. Conversely, every class annihilated by two has this form. These observations prove exactness directly.

Proof of Theorem 2.2. Lemmas 3.1 and 4.1 give $D_B = \ker \bar{\Delta}_B / K$ of dimension two. In weight at most one,

$$\ker \bar{\Delta}_B \cap \bar{\mathcal{F}}_1 = K, \quad (12)$$

because $\bar{\Delta}_B$ equals $\bar{M}_B$ on $\bar{V}_1$.

For $z \in L/\mathcal{F}_2$, reduced modulo two, choose a lift $\tilde{z} \in L/2$ and define

$$\phi(z) = [\bar{\Delta}_B \tilde{z}] \in \ker \bar{\Delta}_B / K \simeq D_B. \quad (13)$$

If the lift changes by $w \in \bar{\mathcal{F}}_2$, its image changes by $\bar{\Delta}_B w \in \bar{\mathcal{F}}_1 \cap \ker \bar{\Delta}_B = K$. Thus the map is well defined without choosing a complement. Its leading weight-two part is $\bar{E}_B z$, which is injective. The map is consequently injective and hence an isomorphism between two-dimensional spaces. Equation (5) identifies its source with $\text{Lie}_3((T/A) \otimes \mathbb{F}_2)$.

Reduction of $F\tau_B F^{-1} = \tau_B^q$ gives $\bar{F}\bar{\tau}_B\bar{F}^{-1} = \bar{\tau}_B$, since q is odd. Thus (13) is equivariant on the kernel quotient. To check equivariance of the Bockstein identification, put $S_q(A) = I + A + \cdots + A^{q-1}$ for $A = \rho_3(\tau_B)$. Then

$$\rho_3(F)\Delta_B = \Delta_B S_q(A)\rho_3(F).$$

Put $G = \rho_3(F)$. For $\bar{x} \in \ker \bar{\Delta}_B$ the identity gives

$$G\beta(\bar{x}) = [G\Delta_B \bar{x}/2] = \beta(\overline{S_q(A)Gx}) = \beta(\bar{G}\bar{x}).$$

The last equality holds because $\bar{G}\bar{x} \in \ker(\bar{A} - I)$ and $S_q(\bar{A})$ acts as one there. Thus the actual Bockstein target action is equivariant. On the highest quotient the action is $\bar{U} \otimes \det \bar{U} = \bar{U}$. The extension block $\bar{H}$ has no effect there. For odd $r \in \mathbb{Z}_2^\times$, write $\Delta_r = A^r - I = \Delta_B S_r(A)$, where

$$S_r(A) = rI + \binom{r}{2}\Delta_B + \binom{r}{3}\Delta_B^2 + \binom{r}{4}\Delta_B^3.$$

This is an integral unit matrix with determinant r^{16} , since $\Delta_B^4 = 0$. It commutes with Δ_B , so their image and kernel are unchanged; also $\bar{\Delta}_r = \bar{\Delta}_B$. In the common cokernel the new Bockstein is $[S_r(A)\Delta_B \bar{x}/2] = r\beta(\bar{x}) = \beta(\bar{x})$, since $S_r(A)$ acts by r on the cokernel and the target is killed by two. The identification is therefore independent of an odd-unit generator change. This proves the primitive assertions. $\square$

5 Even pairings and extension dependence

If $s \geq 1$ then $\bar{\tau}_B = I_T$ and $\bar{\Delta}_B = 0$, so (11) gives $D_B = (L/2)/K$ of dimension $16 - 6 = 10$. If $\bar{U} = I_2$, reduction of (2) is the shear $\bar{\tau}_H$. When $\bar{H} = 0$, the action is trivial.

Lemma 5.1. *If B' and H' are distinct nonzero symmetric matrices over $\mathbb{F}_2$, then the row spaces of $M_{B'}$ and $M_{H'}$ have zero intersection.*

Proof. For parameters (a, b, c) , the three nonzero vectors in the row space are

$$\begin{aligned} r_0 &= (c, 0, 0, 0, a, b), \\ r_1 &= (0, b, c, a, b, c), \\ r_2 &= (c, b, c, a, a + b, b + c). \end{aligned}$$

Each row type determines its parameters injectively. Equating r_0 to a different row type forces the other parameters to vanish by columns 2, 3, 4. Equating $r_1(a, b, c)$ to $r_2(a', b', c')$ gives successively $c' = c = 0$, $b = b'$, $a = a'$, $a' = 0$ and $b' = 0$, using columns 1, 3, 2, 4, 5, 6. Thus no cross-type equality is possible for nonzero parameters. Same-type equality requires the matrices to be equal. $\square$

Proof of Theorem 2.3. For $\bar{H} \neq 0$, put $D = \rho_3(\bar{\tau}_H) - I$. Lemma 4.1 gives $\text{rank } D = 8$ and $\text{im } D = \ker D$. The space K is stable under D , as it is the reduction of the normaliser-stable integral kernel. On $\bar{\mathcal{F}}_1$, the operator D is $\bar{M}_H$ on $\bar{V}_1$ and zero on $\bar{V}_0$, including when an integral lift of H is singular. Since $K \subset \bar{\mathcal{F}}_1$, this gives

$$\dim(\text{im } D \cap K) = 2 + \dim(\ker \bar{M}_H \cap \ker \bar{M}_{B_0}).$$

If $\bar{H} = \bar{B}_0$, the right hand side is six. If the matrices are distinct, Lemma 5.1 makes the stacked map $(\bar{M}_H, \bar{M}_{B_0})$ have rank four, so the right hand side is four. The rank induced by D on $(L/2)/K$ is consequently two or four. Subtracting from ten proves (4). $\square$

6 The complete even-pairing formula

Let $s \geq 1$. In addition to (4), the following describes every compatible graph action modulo two. For a nonzero symmetric form P on a two-dimensional $\mathbb{F}_2$ -space, define

$$\eta(P) = \begin{cases} 1, & P \text{ is nonsingular and nonalternating,} \\ 0, & \text{otherwise.} \end{cases}$$

Theorem 6.1. *If $\bar{U}$ has order three, then $\dim D_B^F = 2$. If $\bar{U}$ has order two, let v be its unique nonzero fixed vector and put $\epsilon = v^t \bar{H} v$. Then*

$$\dim D_B^F = 5 - (2 - \eta(\bar{B}_0))\epsilon. \quad (14)$$

Equivalently, the dimension is five when $\bar{F}^2 = I_T$; when $\bar{F}$ has order four it is four for a nonsingular nonalternating $\bar{B}_0$, and three otherwise. Together with (4), these cases exhaust the possible graph actions.

Proof. All actions are on $(L/2)/K$. Set $D = \bar{U}^{-t}$ and $G = \text{diag}(D, \bar{U})$. Changing the complement by τ_R , with R symmetric over $\mathbb{F}_2$, conjugates $\tau_H G$ to

$$\tau_{H+R+DRD^t} G. \quad (15)$$

This preserves the quotient, since the integral shears commute with monodromy and preserve its kernel. If $\bar{U}v = v$, evaluation of $R + DRD^t$ on v is zero. Thus ϵ is independent of this change.

By congruence there are three nonzero symmetric forms. Their involution stabilisers reduce to the representatives in the following table. In every case (15) reduces H to $\begin{pmatrix} 0 & t \\ t & \epsilon \end{pmatrix}$, with $t \in \mathbb{F}_2$. Substitution in the tensor presentation gives the displayed ranks for either value of t .

$\bar{B}_0$	$\bar{U}$	v	$\text{rank}(F - I)$ if $\epsilon = 0$	if $\epsilon = 1$
$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$	$\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$	$\begin{pmatrix} 0 \\ 1 \end{pmatrix}$	5	7
I_2	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	$\begin{pmatrix} 1 \\ 1 \end{pmatrix}$	5	6
$J = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	$\begin{pmatrix} 1 \\ 1 \end{pmatrix}$	5	7

Appendix A gives the quotient monomials and all twelve row reductions explicitly, so this finite calculation can be checked directly. Subtracting these ranks from ten proves (14). In the same representatives $H + DHD^t = 0$ exactly when $\epsilon = 0$. Since $(\tau_H G)^2 = \tau_{H+DHD^t}$, this also proves the assertion about the order of $\bar{F}$.

An order-three graph action can stabilise only J . Take $\bar{U} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$. On symmetric matrices $R \mapsto R + DRD^t$ has rank two, and its quotient is represented by $0, J$. Indeed $H + DHD^t + D^2H(D^t)^2 = (a + b + c)J$ for $H = (a, b, c)$. The two remaining actions have rank eight on $(L/2)/K$, as the last two explicit row reductions in the appendix show. Their fixed dimension is two. The inverse order-three element is obtained by conjugation. This proves the theorem. $\square$

7 The full two-primary group for a thick theta graph

The subgroup D_B does not determine the full two-primary fixed group. We now compute that group for the pairing used in the eight split examples and their unramified quadratic twists. Write

$$B_\theta = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad B = 2B_\theta, \quad P_B = (\text{coker } \Delta_B)_{\text{tors}}.$$

The torsion here is taken over $\mathbb{Z}_2$. In this section assume $U = \sigma I_2$ integrally, where $\sigma \in \{1, -1\}$; this is stronger than a condition on $\bar{U}$.

Theorem 7.1. *One has $P_B \simeq (\mathbb{Z}/2)^8 \oplus (\mathbb{Z}/4)^2$. For every positive odd q and symmetric H , its Frobenius action is determined by σ and $\bar{H}$. The fixed groups are*

$\bar{H}$	$\sigma = 1$	$\sigma = -1$
0	$(\mathbb{Z}/2)^8 \oplus (\mathbb{Z}/4)^2$	$(\mathbb{Z}/2)^{10}$
$J = \bar{B}_\theta$	$(\mathbb{Z}/2)^8$	$(\mathbb{Z}/2)^6 \oplus (\mathbb{Z}/4)^2$
Other nonzero symmetric forms	$(\mathbb{Z}/2)^4 \oplus (\mathbb{Z}/4)^2$	$(\mathbb{Z}/2)^6$

For tame inertia cohomology the action includes an additional factor q^{-1} . The displayed groups also give its Frobenius invariants when $q \equiv 1 \pmod{4}$, in particular over $\mathbb{Q}_5$.

Proof. Let W be the saturation of $\text{im } \Delta_B$ in L . The weight maps give

$$W = V_0 \oplus V_1 \oplus E_{B_\theta}(V_3).$$

The last summand is saturated because E_{B_θ} has a unit minor. Hence W has rank ten, and $P_B = W/\Delta_B L$. Appendix B displays a basis of this group with generators $x_0, \dots, x_7$ of order two and y_0, y_1 of order four. It also gives an exact image minor and the two lattice inclusions verifying this presentation, without assuming a Smith formula for a general pairing.

Write an element as $\sum \xi_i x_i + \eta_0 y_0 + \eta_1 y_1$, where $\xi_i \in \mathbb{F}_2$ and $\eta_j \in \mathbb{Z}/4$. Put $a = \bar{H}_{00}$, $b = \bar{H}_{01}$, $c = \bar{H}_{11}$ and $\bar{\eta}_j = \eta_j \pmod{2}$. Direct application of the three shears with parameters $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$ gives the following coordinates of $(\rho_3(\tau_H) - 1)$:

$$\begin{aligned} \delta \xi_0 &= a(\xi_3 + \bar{\eta}_0) + b(\xi_7 + \bar{\eta}_1) + c\xi_2 + ac\xi_7, & \delta \xi_3 &= c\xi_7, \\ \delta \xi_1 &= a\xi_5 + b(\xi_6 + \bar{\eta}_0) + c(\xi_4 + \bar{\eta}_1) + ac\xi_6, & \delta \xi_5 &= c\xi_6, \\ \delta \xi_2 &= a\xi_7, & \delta \eta_0 &= 2b\xi_6, \quad \delta \eta_1 = 2a\xi_6 + 2b\xi_7. \\ \delta \xi_4 &= a\xi_6, & & \\ \delta \xi_6 &= \delta \xi_7 = 0, & & \end{aligned}$$

The equations for the ξ_i are in $\mathbb{F}_2$, and those for the η_j are in $\mathbb{Z}/4$. Each generating shear has order two on P_B , so arbitrary integral H reduces to these three parity parameters. The scalar Levi element $\text{diag}(qI_2, I_2)$ acts trivially: the order-four generators have weight one and are multiplied by $q^2 \equiv 1 \pmod{4}$; the other generators are either homogeneous of order two, or a weight-two vector plus twice a weight-one vector. Their displayed expressions give the same conclusion. Finally $\rho_3(-I_T) = -I_L$. Thus F acts as $\sigma \rho_3(\tau_H)$ on P_B .

For $\sigma = 1$, $H = 0$ fixes the whole group. For $H = J$, the equations force $\xi_6 = \xi_7 = 0$ and η_0, η_1 even, leaving eight independent elements of order two. In every other nonzero case $(a, c) \neq (0, 0)$. The equations first force $\xi_6 = \xi_7 = 0$, then impose two independent conditions involving respectively (ξ_2, ξ_3) and (ξ_4, ξ_5) . The two η_j are arbitrary, leaving $(\mathbb{Z}/4)^2 \oplus (\mathbb{Z}/2)^4$.

For $\sigma = -1$ the fixed equation is $\rho_3(\tau_H)z = -z$. The ξ equations are unchanged and the last two equations become $\bar{\eta}_0 = b\xi_6$ and $\bar{\eta}_1 = a\xi_6 + b\xi_7$. For $H = 0$ both η_j are even, giving $(\mathbb{Z}/2)^{10}$. For $H = J$ their parities are ξ_6, ξ_7 , and all remaining equations vanish, giving $(\mathbb{Z}/4)^2 \oplus (\mathbb{Z}/2)^6$. Otherwise $\xi_6 = \xi_7 = 0$, both η_j are even, and the two independent ξ conditions remain, giving $(\mathbb{Z}/2)^6$. This proves the table. The final assertion follows because P_B has exponent four. $\square$

8 Arithmetic realisations over $\mathbb{Q}_5$

For totally toric semistable reduction, Raynaud uniformisation gives an exact sequence of integral Tate modules

$$0 \longrightarrow X^\vee \otimes \mathbb{Z}_2(1) \longrightarrow T_2 J \longrightarrow X \otimes \mathbb{Z}_2 \longrightarrow 0,$$

where the principal polarisation identifies the period lattice with the character lattice $X = H_1(\Gamma, \mathbb{Z})$. Its pairing is the metric graph pairing. These identifications, the component-group formula and graph Galois action are standard; see [4, Sections 2.1–2.2]. The first term is saturated and Lagrangian, so an integral symplectic basis can be chosen with inertia τ_B and arithmetic Frobenius as in (2), with $q = \#k$. The relation with inertia imposes $U^t B U = B$.

The torsion of tame inertia cohomology is computed by the coinvariants, with a Frobenius factor q^{-1} from the choice of tame generator. That factor is one on two-torsion. Wild inertia is trivial on the semistable Tate module at primes different from two and hence on our tensor lattice. This explains the local cohomological interpretation of the calculated Frobenius action. The statements are restricted to odd residue characteristic.

For the theta graph of two vertices joined by three edges of lengths ℓ_1, ℓ_2, ℓ_3 , orient all edges the same way and use cycles $E_1 - E_2, E_2 - E_3$. Its pairing is

$$B = \begin{pmatrix} \ell_1 + \ell_2 & -\ell_2 \\ -\ell_2 & \ell_2 + \ell_3 \end{pmatrix}. \quad (16)$$

The ordinary geometric component group is $\Phi \simeq \text{coker}(B : X \rightarrow X^\vee)$, with action on $X^\vee$ given by U^{-t} . Its Tamagawa number is $\#\Phi^F$, without a cyclotomic factor on this graph formula.

8.1 Primitive examples with three graph actions

Let

$$C_h : \quad y^2 = h(x)^2 - 5, \quad h = x^3 - x, \quad x^3 - 2x, \quad x^3 + x + 1.$$

Each h is square-free modulo five. In the special fibre the two rational components $y = \pm h(x)$ meet at its three geometric roots. Locally, $(y - h)(y + h) = -5$, and the two factors are independent coordinates at a simple root because two and h' are units. Thus all node thicknesses are one. At infinity put $z = 1/x$ and $Y = y/x^3$; the equation has two smooth rational points $Y = \pm 1$ at $z = 0$. These statements give a proper semistable genus-two model with theta graph and

$$B_\theta = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

The node residue degrees are respectively $(1, 1, 1)$, $(1, 2)$ and (3) . Frobenius preserves each component $y = \pm h(x)$. With the rational node labelled E_2 in the middle case, its action interchanges E_1 and E_3 , and hence sends the two cycles to the negatives of each other. The induced graph actions are therefore I_2 , $\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$ and, after labelling the cycle, $\begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}$. Theorem 2.2 gives the last column below.

h	Node degrees	Ordinary Tamagawa	$\dim D_{C_h}^F$
$x^3 - x$	$(1, 1, 1)$	3	2
$x^3 - 2x$	$(1, 2)$	3	1
$x^3 + x + 1$	(3)	3	0

Here $\Phi \simeq \mathbb{Z}/3$ and U^{-t} acts trivially in all three cases. Thus all three ordinary Tamagawa numbers agree while the cubic fixed dimensions differ. Exact discriminants of the three sextics are nonzero with five-adic valuation three, as recorded in the supplement.

8.2 Identical graphs with different cubic invariants

Theorem 8.1 (Eight-curve realisation). *There are eight explicit genus-two curves over $\mathbb{Q}_5$ with the same split metric theta graph of edge lengths $(2, 2, 2)$, trivial graph Frobenius, geometric component group $\mathbb{Z}/2 \oplus \mathbb{Z}/6$ and ordinary Tamagawa number twelve, but for which the Frobenius-fixed dimensions of the cubic two-torsion are respectively 10, 8 and 6. In particular, the metric reduction graph equipped with its Frobenius action does not determine the integral cubic invariant in the even case.*

Proof. For $u_i \in \{1, 2\}$, let

$$C_{u_1, u_2, u_3} : y^2 = (x^2 - 25u_1)((x-1)^2 - 25u_2)((x+1)^2 - 25u_3). \quad (17)$$

All eight special fibres are $y^2 = x^2(x-1)^2(x+1)^2$. At a centre $a \in \{0, 1, -1\}$ the product of the other two factors is a unit whose residue is a square. Hensel's lemma supplies a local square root. Dividing y by it gives

$$y'^2 = (x-a)^2 - 25u_i, \quad (y' - (x-a))(y' + (x-a)) = -25u_i.$$

Each node has thickness two and split branches over $\mathbb{F}_5$. Resolving each node inserts a rational component, so every model has the same split metric theta graph with lengths $(2, 2, 2)$. Its pairing is $2B_\theta$, its geometric component group is $\mathbb{Z}/2 \oplus \mathbb{Z}/6$, and its ordinary Tamagawa number is twelve. The infinity chart is smooth with two rational points as above. The exact sextic discriminants are nonzero and have valuation six.

Label the roots by three pairs $\{r_0, r_1\}$, $\{r_2, r_3\}$, $\{r_4, r_5\}$ in the order of the factors. The standard description of $J[2]$ is the space of even subsets of these six roots modulo the full set, with pairing given by intersection cardinality modulo two. The classes

$$w_0 = \{0, 1\}, \quad w_1 = \{2, 3\}, \quad \beta_0 = \{1, 4\}, \quad \beta_1 = \{0, 1, 3, 4\}$$

are a symplectic basis. The three within-pair transpositions act as shears $v_i v_i^t$ on this basis, where

$$v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad v_3 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

The pair classes are the reductions of the vanishing cycles. Their first two span the toric Lagrangian, so this symplectic basis can be lifted to an adapted integral basis.

If $u_i = 1$ the roots of that factor are rational over $\mathbb{Q}_5$. If $u_i = 2$, they lie in its unramified quadratic extension and Frobenius interchanges them. Graph Frobenius remains trivial in every case. Putting $\epsilon_i = 0$ or 1 according as $u_i = 1$ or 2, we therefore have

$$\bar{U} = I_2, \quad \bar{H} = \sum_{i=1}^3 \epsilon_i v_i v_i^t, \quad \bar{B}_\theta = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Choosing no pair transpositions gives $\bar{H} = 0$; choosing all three gives $\bar{H} = \bar{B}_\theta$; the remaining six choices give the six other nonzero symmetric matrices. Hence

Units	Number of curves	$\dim D_C^F$	$\#D_C^F$
$(1, 1, 1)$	1	10	1024
$(2, 2, 2)$	1	8	256
All other triples in $\{1, 2\}^3$	6	6	64

The table realises all three cases of Theorem 2.3. This proves the claimed failure of the metric graph, even equipped with its Frobenius action, to determine the cubic integral invariant in the even case. The statement does not assert equality of the full integral Tate modules of these curves. $\square$

8.3 Nontrivial graph actions and identical ordinary comparisons

For $m \in \{2, 4\}$ and $v(x) \in \{1, x\}$, consider

$$C_{m,v}: y^2 = (x^2 - 5^m)((x^2 - 2)^2 - 25v(x)). \quad (18)$$

The special fibre is $y^2 = x^2(x^2 - 2)^2$. Its two components are individually fixed, with one rational node at zero and a conjugate pair at $\pm\alpha$, where $\alpha^2 = 2$ in $\mathbb{F}_{25}$. Their thicknesses are $(m, 2, 2)$: after removing the square unit from the other factor, the local equations have smoothing parameters of valuations $m, 2, 2$. The infinity chart is smooth. The exact discriminants are nonzero and have valuations $m + 4$.

For $v = 1$, both nodes over $\mathbb{F}_{25}$ have square smoothing units, so Frobenius interchanges the two root pairs without an internal twist. For $v = x$, the relevant unit is α , with norm $-2 = 3$ to $\mathbb{F}_5$, a nonsquare. Its square root lies in the unramified degree-four extension. Frobenius therefore acts by a four-cycle on these four roots. The roots above zero are rational in both cases.

In the root-subset basis used above, the respective permutations may be labelled $(2\ 4)(3\ 5)$ and $(2\ 4\ 3\ 5)$. They give $\bar{U} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$ and $\bar{H} = (0, 1, 0)$ or $(0, 1, 1)$, respectively. Thus $\epsilon = 0$ or 1 . Using cycles $E_0 - E_2, E_1 - E_2$, the pairing and signed graph action are

$$B = \begin{pmatrix} m+2 & 2 \\ 2 & 4 \end{pmatrix}, \quad U = \begin{pmatrix} 1 & 0 \\ -1 & -1 \end{pmatrix}.$$

They do not depend on v . For $m = 2$, $\bar{B}_0$ is alternating; for $m = 4$, it is nonsingular and nonalternating. Theorem 6.1 gives

m	v	Ordinary geometric Φ	Ordinary Tamagawa number	$\dim D_C^F$
2	1	$\mathbb{Z}/2 \oplus \mathbb{Z}/6$	6	5
2	x	$\mathbb{Z}/2 \oplus \mathbb{Z}/6$	6	3
4	1	$\mathbb{Z}/2 \oplus \mathbb{Z}/10$	10	5
4	x	$\mathbb{Z}/2 \oplus \mathbb{Z}/10$	10	4

For each m the two curves have identical metric graphs, signed graph Frobenius and ordinary component groups with action. The different cubic dimensions measure their integral extension data. Finally $y^2 = (x^3 + x + 1)^2 - 25$ has three conjugate thickness-two nodes, order-three graph Frobenius, ordinary Tamagawa number three and cubic fixed dimension two.

8.4 Full groups and unramified quadratic twists

Theorem 7.1 upgrades the original eight split examples to full two-primary fixed groups. Twisting each by the nonsquare unit two multiplies Tate Frobenius by -1 and hence multiplies its cubic action by -1 . It leaves the six-root action on $J[2]$ unchanged. The graph action becomes $U = -I_2$: the two components are interchanged while the three nodes remain rational. These eight twists share the same metric graph and graph action. Their ordinary component group is still $\mathbb{Z}/2 \oplus \mathbb{Z}/6$, with Tamagawa number four, since its fixed subgroup is $\Phi[2]$.

Each twisted sextic is twice the original one, so its discriminant is 2^{10} times the original discriminant and again has valuation six. At infinity the two smooth geometric points are conjugate in the unramified quadratic extension. For the two families the full groups are

Unit triple	Split family	Unramified quadratic twist
$(1, 1, 1)$	$(\mathbb{Z}/2)^8 \oplus (\mathbb{Z}/4)^2$	$(\mathbb{Z}/2)^{10}$
$(2, 2, 2)$	$(\mathbb{Z}/2)^8$	$(\mathbb{Z}/2)^6 \oplus (\mathbb{Z}/4)^2$
Other six triples	$(\mathbb{Z}/2)^4 \oplus (\mathbb{Z}/4)^2$	$(\mathbb{Z}/2)^6$

All groups refer to the specified cubic lattice. Since $5^{-1} \equiv 1 \pmod{4}$, the tame-cohomology twist does not change this table. In particular, in the split family the last two rows have the same full order, 256, but different group structures. In the twisted family the first two rows have the same full order, 1024, but different group structures. Conditions on $\bar{U}$ alone cannot replace the integral $U = \pm I_2$ hypothesis in this calculation.

9 Relation to existing work and computational verification

The surface Lie presentation is classical [1, 2]. Berry’s Proposition 2.8 explicitly constructs the tensor lattice (1) and proves its saturation in a broader hyperelliptic setting [3]; the present work does not claim that lattice or the cubic Chabauty criterion as a contribution. The ordinary graph pairing and component-group calculations are also established [4, 5].

Betts–Dogra give a graph description of rational unipotent fundamental groupoids and local Kummer maps [6], while Betts–Litt study rational weight-monodromy and Frobenius semisimplicity [7]. The integral two-torsion retained here is lost after tensoring with $\mathbb{Q}_2$, so the examples are complementary to those rational results. The proposed contribution is the explicit integral Bockstein identification (3), the complete even fixed-dimension formula, the full thick-theta fixed groups and their arithmetic realisation. The reduced sixteen-dimensional symplectic representation is classical: Chastkofsky–Feit [8, Theorem 3.2] records the corresponding Steinberg module. Appendix C gives an explicit intertwiner with building homology and a regular orbit basis along the Siegel radical. Its two-dimensional Levi coinvariant is likewise consistent with the standard parabolic theorem recalled in [9, Lemma 9.10]. These modular facts are background, not proposed contributions; the calculation tracks the integral kernel and its Bockstein quotient.

The accompanying file `verify_cubic_torsion.py` uses Python’s standard library and constructs the full twenty-four-coordinate tensor presentation before passing to a unimodular quotient basis. It verifies (9) coefficient by coefficient over $\mathbb{Z}[a, b, c]$, checks the determinant polynomial, all seven nonzero symmetric parities, all 288 compatible finite normaliser actions, and the forty-two distinct row-space pairs. In its primitive quotient basis the isomorphism (13) makes the action equal to $\bar{U}$ directly. The twenty-four sextic discriminants are calculated by both a fraction-free Sylvester determinant and rational Euclidean resultants. Fourteen normal-form row reductions give the complete even formula. For the full thick-theta torsion the code supplies exact lattice and action witnesses, checks 512 normalisers, and separately tests all 4096 elements for each of sixteen sign/parity actions. It constructs the building comparison from its thirty vertices and forty-five edges and verifies the full 720-element generated group. The verifier is self-contained.

A second exact calculation constructs the free cubic Lie lattice inside the sixty-four associative words of length three, using twenty Lyndon brackets and four primitive surface relations with a unimodular completion. The cubic action is reconstructed by exact polynomial interpolation, justified by its degree bound, and compared entry by entry with the derivation calculation. All 256 entries of (9) agree.

Integer Smith reduction with both unimodular factors checks 28 physical Bocksteins, all 288 finite cases, fourteen printed reductions, 448 actual normaliser-equivariance cases, 56 odd-generator changes and 108 nonsingular signed pairings. For the full group it checks 512 normalisers and enumerates 65536 elements; a further 65536 element tests calculate the sixteen split/twisted curve groups directly in the root-adapted basis.

For all twenty-four curves, lifted roots determine Frobenius before extracting $\bar{H}$. Critical points and unit Hessians determine the smoothing constants, and a six-by-six Bezoutian recalculates each integer discriminant. Root distances give a separate valuation check. Appendix D

supplies the arguments converting these finite calculations to exact geometric statements. The supplement retains the transformations and local witnesses.

The computational checks are intended as reproducibility support for the displayed arguments and do not replace the proofs.

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A Explicit finite row reductions

Use the monomial basis below, which is the integral basis of Section 3 in another order:

$$\begin{aligned}
 z_0 &= b_{0;01}, & z_1 &= b_{0;03}, & z_2 &= b_{1;01}, & z_3 &= b_{1;02}, \\
 z_4 &= b_{1;03}, & z_5 &= b_{1;12}, & z_6 &= b_{2;01}, & z_7 &= b_{2;02}, \\
 z_8 &= b_{2;03}, & z_9 &= b_{2;12}, & z_{10} &= b_{2;23}, & z_{11} &= b_{3;01}, \\
 z_{12} &= b_{3;02}, & z_{13} &= b_{3;03}, & z_{14} &= b_{3;12}, & z_{15} &= b_{3;23}.
 \end{aligned}$$

For the three normal forms (a, b, c) the following monomial classes form ordered bases of $(L/2)/K$:

(a, b, c)	Indices of the ten quotient monomials z_i
$(1, 0, 0)$	5, 6, 7, 8, 9, 10, 12, 13, 14, 15
$(1, 0, 1)$	1, 4, 7, 8, 9, 10, 12, 13, 14, 15
$(0, 1, 0)$	3, 7, 8, 9, 10, 11, 12, 13, 14, 15

To specify a row of a matrix on ten coordinates, encode $(r_0, \dots, r_9)$ by the integer $\sum_{j=0}^9 r_j 2^j$. For $H = (0, t, \epsilon)$ and the involutions in the proof of Theorem 6.1, the nonzero rows of the reduced row echelon form of $F - I$ have the following codes:

$\bar{B}_0$	t	ϵ	Nonzero row codes
100	0	0	1, 4, 328, 16, 32
100	0	1	65, 4, 72, 16, 32, 256, 512
100	1	0	769, 516, 840, 16, 32
100	1	1	65, 4, 72, 16, 32, 256, 512
101	0	0	3, 68, 264, 144, 544
101	0	1	259, 68, 264, 144, 32, 512
101	1	0	451, 580, 776, 144, 544
101	1	1	195, 68, 264, 144, 32, 512
010	0	0	33, 66, 260, 136, 528
010	0	1	161, 386, 260, 136, 16, 448, 512
010	1	0	545, 578, 772, 136, 528
010	1	1	161, 386, 260, 136, 16, 448, 512

The labels 100, 101, 010 abbreviate (a, b, c) . For $B_0 = 010$, $U = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$ and either $H = 0$ or $H = 010$, the eight codes are

$$1, 2, 388, 136, 16, 32, 64, 512.$$

These tables can be obtained by applying the tensor action to the ten monomials, reducing with the eight integral relations, and then eliminating $K = V_0 \oplus \ker M_{B_0}$. The rows give the ranks used in the proof, rather than a sample of pairings.

B An exact presentation of the full thick-theta torsion

For $B = 2B_\theta$, the following classes in $W/\Delta_B L$ give the cyclic basis used in Theorem 7.1:

$$\begin{aligned}
x_0 &= 3z_0 + 2z_2, & x_1 &= 3z_2, \\
x_2 &= 3z_1 + 2z_4 + 2z_{11}, & x_3 &= 3z_3 + 2z_4 + 3z_6 + 2z_{11}, \\
x_4 &= 3z_4 + 2z_6, & x_5 &= 3z_5 + 2z_6, \\
x_6 &= 2z_6 + 3z_7 + 4z_8 - 2z_9 - 2z_{12} + z_{14}, \\
x_7 &= -2z_7 - z_8 + 2z_{11} + 3z_{12} + 2z_{13} - 4z_{14}, \\
y_0 &= z_6 + 2z_{11}, & y_1 &= 3z_{11}.
\end{aligned}$$

The eight x_i have order two and the two y_j have order four. Here is a precise lattice check of this assertion and generation. Order a basis of W by the bases of V_0, V_1 in Section 3 followed by $E_{B_\theta}(z_{10}), E_{B_\theta}(z_{15})$. Let C be the 10×16 matrix of $\Delta_B : L \rightarrow W$, and let G be the 10×10 matrix of the ten displayed generators in W . Direct substitution in (9) or direct application of $\rho_3(\tau_B)$ gives

$$\det G \in \mathbb{Z}_2^\times, \quad G^{-1}C = \text{diag}(2, 2, 2, 2, 2, 2, 2, 2, 4, 4)Z, \quad Z \in M_{10 \times 16}(\mathbb{Z}_2).$$

Conversely, the ten columns of $G \text{diag}(2, \dots, 2, 4, 4)$ belong to $C\mathbb{Z}_2^{16}$. These two inclusions prove

$$C\mathbb{Z}_2^{16} = G \text{diag}(2, \dots, 2, 4, 4)\mathbb{Z}_2^{10}.$$

For an additional exact check, the image minor in zero-based domain columns

$$3, 7, 8, 9, 10, 11, 13, 15, 12, 14$$

has determinant $110592 = 27 \cdot 2^{12}$. Expressing every column of C in these ten columns has odd denominators. The standalone certificate supplies the full rational coordinates, relation witnesses and shear-action witnesses, recording every rational number as its numerator and denominator. The original Smith calculation is performed modulo 2^8 , with all elementary transformations checked; the exact lattice inclusions do not depend on numerical approximation. The independent word-basis calculation also gives the ordinary integer Smith diagonal

$$2, 2, 2, 2, 2, 2, 2, 2, 12, 36, 0, 0, 0, 0, 0, 0.$$

Integral unimodular matrices P, Q satisfy $P\Delta_B Q$ equal to this diagonal. Localising at two removes the odd parts of 12 and 36, giving $(\mathbb{Z}/2)^8 \oplus (\mathbb{Z}/4)^2$.

C Comparison with the classical Steinberg module

The symplectic building of $\mathbb{F}_2^4$ has fifteen nonzero vectors as points, fifteen totally isotropic planes as lines, and forty-five incidences. Its incidence graph has thirty vertices, is connected, and has first homology of dimension sixteen. Its top homology is the usual Steinberg module.

An explicit comparison avoids relying only on matching dimensions or characters. Sort the points by their binary integers $1, \dots, 15$, with bit i giving the e_i coordinate. Sort the isotropic planes lexicographically by their triples of nonzero vectors, and sort edges (p, plane) lexicographically. The greedy spanning tree has twenty-nine edges; its non-tree edge indices are

$$22, 23, 25, 26, 28, 29, 31, 32, 34, 35, 37, 38, 40, 41, 43, 44.$$

For each non-tree edge take its fundamental cycle. In that homology basis and the monomial basis (z_i) above, an intertwiner $X : L/2 \rightarrow H_1$ has rows encoded by the following sixteen integers, using sixteen bits:

$$\begin{aligned} &4, 36, 1228, 1772, 10260, 59444, 48604, 31740, \\ &5, 63, 613, 1503, 10263, 55341, 61303, 40141. \end{aligned}$$

The matrix has rank sixteen. For each of the fifteen integral symplectic transvections $I + v(v^t\Omega)$, where v runs over the nonzero binary vectors and Ω is the symplectic matrix, direct multiplication gives $X\rho_3(g) = g_*X$ modulo two. These transvections generate all 720 elements of $\text{Sp}_4(\mathbb{F}_2)$. Thus this is a finite, explicit module isomorphism, not a claim that the representation is new.

There is also an elementary check along the Siegel unipotent radical $N = \{\tau_H : H \in \text{Sym}_2(\mathbb{F}_2)\}$ of order eight. The sixteen vectors $\rho_3(\tau_H)z_{10}, \rho_3(\tau_H)z_{15}$ form an orbit basis. Hence $L/2$ is a free $\mathbb{F}_2[N]$ -module of rank two, and its augmentation image is $\bar{\mathcal{F}}_2$ of dimension fourteen. Its N -coinvariant quotient is therefore exactly $\text{Lie}_3(\bar{\Lambda})$, the standard two-dimensional Levi module. This agrees with classical parabolic Steinberg theory. The primitive Bockstein isomorphism in the main text additionally identifies this quotient with $\ker \bar{\Delta}_B/K$ using the integral kernel and the map $z \mapsto \bar{\Delta}_B z$.

D Geometric and arithmetic hypotheses in the examples

The following details make explicit the geometric inputs used in Section 8. They also explain why finite-precision computations of critical points and root permutations suffice for the examples.

Lemma D.1 (Proper models and node thicknesses). *Let R be a complete discrete valuation ring of odd residue characteristic and let $f \in R[x]$ have degree six with unit leading coefficient c . Suppose $\bar{f} = \bar{c}\bar{h}^2$, where $\bar{h}$ is a square-free cubic. Suppose also that f has nonzero discriminant. For each root s of $\bar{h}$, let c_s be the unique lift near s satisfying $f'(c_s) = 0$, over the corresponding unramified extension. If $a_s = f(c_s) \neq 0$, the completed geometric local model has equation*

$$uv = a_s.$$

Its node thickness is $v(a_s)$. The resulting proper stable model has two rational components joined at the three geometric nodes. Resolving a thickness- ℓ node inserts $\ell - 1$ rational components and gives an edge of metric length ℓ .

Proof. Glue the affine chart $y^2 = f(x)$ to the chart $Y^2 = z^6 f(1/z)$ by $z = 1/x$ and $Y = y/x^3$. This is the finite double cover of $\mathbb{P}_R^1$ defined using $\mathcal{O}(3)$, so it is proper and flat. Its generic fibre is smooth of genus two. At infinity, $\bar{Y}^2 = \bar{c}$ and the derivative $2\bar{Y}$ is a unit. Away from the roots of $\bar{h}$ the derivative in y is a unit. The only geometric special-fibre singularities are consequently the three nodes.

At a root s , one has $\bar{f}''(s)/2 = \bar{c}\bar{h}'(s)^2 \neq 0$. Hensel's lemma gives c_s , and Taylor expansion gives

$$f(x) = a_s + (x - c_s)^2 u_s(x), \quad u_s(c_s) = f''(c_s)/2 \in R^{\text{sh}, \times}.$$

In the strict henselian completion, choose a square root of u_s and set $t = (x - c_s)\sqrt{u_s(x)}$. This is a local coordinate with unit derivative. The coordinates $u = y - t$, $v = y + t$ give the asserted equation, since two is a unit. Its two special-fibre branches are distinct. The normalisations of the components $y = \pm\sqrt{\bar{c}\bar{h}}$ are copies of $\mathbb{P}^1$, each with three attachment points, so the model is stable.

After absorbing a constant unit, a thickness- ℓ equation is $uv = \pi^\ell$. Blowing up (u, π) has one chart $u = \pi u_1$, with remaining equation $u_1 v = \pi^{\ell-1}$. The other chart is regular. Repeating gives the stated chain and length. Thus a stable thickness greater than one is not mistaken for a regular total-space point. $\square$

For all twenty-four examples, the independent verifier lifts c_s , checks the unit Hessian and calculates $v(f(c_s))$. It retains arithmetic internally modulo 5^9 and outputs roots and smoothing data modulo 5^5 . The greatest thickness is four, so all valuations are detected before the precision boundary. This applies in particular to $v(x) = x$ in (18), where a smoothing unit depends on the local coordinate. The critical-point argument removes that dependence before assigning a thickness.

D.1 Root-pair classes and the integral graph basis

Write w_e for the even-subset class of the two roots colliding at node e . In a one-node hyperelliptic smoothing, exchanging those two roots acts by

$$x \mapsto x + \langle x, w_e \rangle w_e$$

on $J[2]$. The integral Picard–Lefschetz formula [10, Exposé XV, §3] identifies this with the reduction of the transvection in the vanishing cycle δ_e . A nonzero transvection determines its vector over $\mathbb{F}_2$, hence $\bar{\delta}_e = w_e$. This uses the smoothing parameter before pulling it back to a power of the uniformiser. An even thickness changes the monodromy power, and does not multiply the generator of the saturated vanishing-cycle lattice by two.

For the three-node theta degeneration these vanishing cycles generate A , with primitive relation $\delta_0 + \delta_1 + \delta_2 = 0$, after compatible orientations. Choose cycles $E_0 - E_2, E_1 - E_2$ and the dual vanishing-cycle basis. The pair classes w_0, w_1 are its reduction modulo two. The

integral vanishing-cycle exact sequence has node-to-component map $\mathbb{Z}_2^3 \rightarrow \mathbb{Z}_2^2$ with all columns $(1, -1)^t$. Its kernel is the cycle lattice; by duality the vanishing-cycle lattice is $\mathbb{Z}_2^3/\mathbb{Z}_2(1, 1, 1)$. This quotient is free, so δ_0, δ_1 form a basis of the saturated toric lattice. The pairing is

$$B_+ = \begin{pmatrix} \ell_0 + \ell_2 & \ell_2 \\ \ell_2 & \ell_1 + \ell_2 \end{pmatrix}.$$

For equal edge lengths, changing the second cycle and its dual by a sign gives convention (16). This integral change is the identity modulo two, so it leaves the root-subset basis and all $\bar{H}$ unchanged. It also leaves $U = \pm I_2$ unchanged. For the four curves (18), B_+ is already the displayed pairing; exchanging nodes 1, 2 while fixing the components gives the displayed signed U .

The even-subset basis in Theorem 8.1 is symplectic for the intersection pairing. If a root permutation gives

$$\bar{F} = \begin{pmatrix} \bar{U}^{-t} & V \\ 0 & \bar{U} \end{pmatrix},$$

the extension matrix is $\bar{H} = V\bar{U}^{-1}$. In particular, V alone must not be substituted for $\bar{H}$. The independent calculation derives the permutations from lifted roots and checks this decomposition. For the two permutations in (18), it gives respectively $\bar{H} = (0, 1, 0)$ and $(0, 1, 1)$.

D.2 Why the root calculations determine Frobenius

The unramified rings used in the fresh verifier are $\mathbb{Z}/5^9[t]/(g)$ for explicitly checked irreducible $\bar{g}$ of degrees one to four. Frobenius sends t to the unique root of g congruent to t^5 modulo five. At a node, the polynomial is evaluated after $x = c_s + 5^{\ell/2}T$ and divided by 5^ℓ . Its two residue roots have unit derivatives. Division loses at most four digits, and the original coordinate is then retained at least modulo 5^5 . The residue solutions therefore determine exact Hensel branches. Distinct roots have distance valuation at most two, strictly below the output precision, so a Frobenius match identifies an exact branch.

For the thickness-one examples, further adjoin $\pi^2 = 5$ and use $x = c_s + \pi T$. Frobenius is chosen to fix π , a legitimate arithmetic Frobenius lift. Replacing the lift by inertia does not change its action on the coinvariants. The ramified involution on $J[2]$ is also checked against the reduced monodromy pairing.

D.3 The cohomological normalisation

Let c be a tame cocycle and put $z = c(\tau)$ for the chosen generator. Arithmetic Frobenius acts by

$$(F \cdot c)(\tau) = \rho_3(F)c(F^{-1}\tau F).$$

Since $F^{-1}\tau F = \tau^{q^{-1}}$, evaluation modulo coboundaries gives $q^{-1}\rho_3(F)[z]$ in the coinvariants. This proves the extra factor in Section 7. It is one on D_B , and on the full exponent-four group when $q \equiv 1 \pmod{4}$. For $q = 3$, $U = I_2$, $H = 0$, the untwisted full invariant group has order 4096, whereas the cohomology invariant group is $(\mathbb{Z}/2)^{10}$ of order 1024. The congruence condition on q is therefore necessary for the displayed cohomology table.